Side A: the row `ic_online_indexbuild2_r770`
source 4008a47a… (PR #1009's frozen digest, re-derived)
rank stage, frame loop: per state 83 shifts × 2 roots × 2 partners
key of every partner: `lift_scan_wide(v, 83)`, then prefetch
probe loop: lookups in key order, exact-order resolution at a hit

→

Side B: rank finder `ic_online_rankfinder_r770`
source 301615b3… = 4008a47a… + 6 counted edits (invertible)
same partner words, same order, same candidates and pair starts
keys: `lift_linear_lanes8::<83>` (round 2's C7) per 8 pending,
tail ≤ 7 by the parent's call; the probe loop is the parent's text

Same output, gated per row and per seed: per M38 rank seed the same rank relation digest, per-attempt probe vector, root calls and counters across kM38IC, kM38RF, kM38RFaa and kM38RFX; rank tables equal to the frozen verified table; one index and layout digest set per base (K600, M38, M10); Q0 on all three bases with both binaries; one relation, probe count and scalar per point; every row's `rank_probe_pipeline` its arm's.

O1 (Q0 exactness)
cPIC, cPRF, cM10IC, cM10RF,
cM38IC, cM38RF (shuffled)

P1 … P12: one segment per new held-out point (role `rankfinder-cold-v1`),
four sub-blocks in a seeded order (`random.Random(2026102170 + i)`):
primary: 3 pairs, sides ABABAB or BABABA; a side = rank process (seed $1800+3i+j$) + cold process
A/A: 2 pairs B vs Baa (kM38RFaa + cM38RFaa), seeds of primary pairs 1, 2
planted: kM38RF vs kM38RFX (frame_plant), seed 7, alternating
rho: one rT, seed $1050000+$ point seed

Primary: $R = \sum E2E(A) / \sum E2E(B)$ over 36 pairs, $E2E$ = rank stage (rank process) + cold (cold process); two-stage paired bootstrap (points, then pairs), 10,000 draws. **Improvement iff the 95% lower bound** > 1. Stage secondary (labelled): rank stage kM38IC / kM38RF. Controls: A/A E2E and cold in $[0.97, 1.03]$; planted rank-stage lower bound > 1. 282 processes on one pinned CPU of one GitHub `ubuntu-24.04` runner; time guard 165 min, job limit 180 min. Any failed or unknown gate, missing row or time-guard skip: **invalid**.